BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, DI KHAN

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2022

PHYSICS

Past Paper 2022 • Paper Code: DIK-SSC-X-2022 (A)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION – A

Time Allowed: 20 min | **Maximum Marks:** 12 | **Paper Code:** DIK-SSC-X-2022 (A)

Q1. Choose the correct option and fill in empty box, given against each statement:

Q1. The resistance of wire will decrease by increasing_____

(A) Temperature
(B) Length
(C) Diameter
(D) Both A and B

Q2. Which of the given have highest ionization power?

(A) Alpha radiation
(B) Beta radiation
(C) Gamma radiation
(D) Delta radiation

Q3. Software that allows users to interact with the computer system is _____

(A) Hardware
(B) Operating system
(C) Application
(D) Program

Q4. Radium-226 decays by emitting an alpha particle. What is the daughter nucleus?

(A) Rd
(B) Bi
(C) Rn
(D) Pb

Q5. The frequency of oscillation for time period 0.025.

(A) 1/50 Hz
(B) 1/5 Hz
(C) 50 Hz
(D) 500 Hz

Q6. The speed of sound on a warm day when the outdoor temperature is 38^0C is _____

(A) 331 m/s
(B) 345 m/s
(C) 351 m/s
(D) 354 m/s

Q7. Minimum echo distance is reduce in _____

(A) Winter (B) Summer

(C) Space (D) Spring

Q8. Prolonged exposure to sounds of about _____ dB can begin to damage hearing.

(A) 50 (B) 60

(C) 85 (D) 120

Q9. A convex lens with focal length 8cm has the power of lens:

(A) 2.05D (B) 12.5D

(C) 12.5D (D) 16.0D

Q10. The value of coulomb constant K, depends on _____

(A) Value of charges (B) Separation of charges

(C) Material medium (D) All of these

Q11. The SI unit of electric field intensity is _____

(A) NC^2 (B) NC^{-1}

(C) Nm (D) NA

Q12. When the current through a wire is 2.0 A and voltage across its end is 3.0 V then resistance_____

(A) 1.5Ω (B) 2.5Ω

(C) 6Ω (D) 4Ω

SECTION – B

Time Allowed: 2:40 Hrs | **Maximum Marks:** 21 | **Paper Code:** DIK-SSC-X-2022 (A)

Q2. Attempt any 8 questions of the following:

(i) Prove the relation between wave speed, wavelength and frequency of wave.

(ii) Why does your voice sound fuller in the shower?

(iii) What are optical fibers? Give three applications of optical fibers?

(iv) Calculate the speed of light in zircon with index of refraction n = 1.923, a material used in Jewelry to replicate diamond.

(v) How does shuffling feet across a carpet cause hair to stand on your body?

(vi) What is potential difference and emf? Differentiate between them.

(vii) Write a short note on Fleming's Left hand rule. Draw the diagram.

(viii) State Joul's Law and write the mathematical form.

(ix) Describe the operation of AND, NAND logic gates by drawing their symbols and truth tables.

(x) What is email and internet? List two uses of internet in daily life.

(xi) What are radio isotopes? Explain their three uses for various applications.

SECTION – C

Maximum Marks: 24

> Note: Attempt any 3 questions: Each carry 8 marks.

Q3.

(a) What is Simple Harmonic Motion (SHM)? What are the conditions for an object to oscillate with SHM?

(b) What is the speed of sound in air at $-20^{o}C$?

Q4.

(a) What is total internal reflection? How we can calculate the critical angle for total internal reflection? What are the conditions for total internal reflection?

(b) What is capacitor? Define capacitance and its units.

Q5.

(a) What is series combination of resistors? How we can determine equivalent resistance for different resistors connected in series?

(b) Describe the phenomena of electromagnetic induction. List the factors effecting electromagnetic induction.

Q6.

(a) Radio-active sources are said to have half-life. What is the meaning of half-life?

(b) Lead-210 has a half-life of 22.3 years. How much of the 80mg of lead will be left after 66.9 years?